

MACRO L3 - 2014-2015

Lecture 7

F. PORTIER

Session 11

These notes are those I wrote during class. They are posted "for the record".

Session 11

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CACAN'S MODEL

Money demand $m_t + v_t = p_t c_t$

$$v(i_t) = 1 - \alpha i_t$$

i_t = nominal interest rate
= opportunity cost of holding
money

Fisher equation (arbitrage equation)

$$i_t = R_t + \pi_t^e$$

$$\pi_t^e = P_{t+1}^e - P_t = \text{expected inflation}$$

exaple

Example

1

2

lend 1 apple $\rightarrow$ receive 1.5 apples
tomorrow

$\hookrightarrow$ financial contract in real terms

$$r = 50\% / \text{day}$$

Assume price of an apple = 1 €
today

Nominal contract

1	2
$1 \in$	$1.5 \times p_2 \in$

$$1 + i^{\circ} = (1.5) \times \frac{p_2}{p_1} e$$

nominal interest rate

$\pi = \frac{p_2 - p_1}{p_1} = \text{inflation rate}$

$(1 + \pi)^e$

Fisher equation

$$(1+i) = (1+r)(1+\pi^e)$$

$\forall i, r, \pi^e$ are small

$$\implies i = r + \pi^e$$

$$\begin{cases} m_t + v_t = p_t + c_t \\ v_t = +\alpha \dot{i}_t \\ \dot{i}_t = n_t + \pi_t^e \end{cases}$$

$$m_t - p_t = c_t - \alpha \dot{i}_t$$

real balances

$$m_t - p_t = c_t - \alpha (n_t + \pi_t^e)$$

$$m_t - p_t = c_t - \alpha r_t - \alpha \pi_t^e \quad (*)$$

• Simplifying assumptions:

$$\text{short run: } \begin{cases} c_t = c \\ r_t = r \end{cases}$$

• normalization

$$(*) \Rightarrow \boxed{m_t - p_t = -\alpha \pi_t^e} \quad \begin{matrix} \text{money} \\ \text{demand} \end{matrix}$$

$c = r = 0$

$$m_t - p_t = -\alpha \pi_t^e$$

↳ I also need a theory for expectations.

$$\pi_t^e ? \quad (= p_{t+1}^e - p_t)$$

Now I assume adaptive expectations.

$$\pi_t^e = \lambda \pi_{t-1}^e + (1-\lambda) \pi_{t-1} \quad \lambda \in [0,1]$$

or equivalently:

$$\pi_t^e = \pi_{t-1} + \lambda (\pi_{t-1}^e - \pi_{t-1})$$

error

yesterday forecast error

- if $\lambda = 0$, naive expectations:

- if $\lambda > 0$: "error correcting mechanism"
 $|\lambda| < 1$

The model has 2 eq:

$$\begin{cases} (1) & m_t - p_t = -\alpha \pi_t^e = -\alpha (p_{t+1}^e - p_t) \\ (2) & \pi_t^e = \lambda \pi_{t-1}^e + (1-\lambda) \pi_{t-1} \end{cases}$$

+ $m_t = \text{money supply} = \text{exogenous}$

unknowns : $\left. \begin{matrix} p_t \\ p_{t+1}^e \end{matrix} \right\} \forall t$

= Theory of the price level and inflation

Soln. Assume that the solution is
of the type

$$P_t = \beta_1 P_{t-1} + \beta_2 m_t + \beta_3 m_{t-1}$$

and let us find $\beta_1, \beta_2, \beta_3$

$$(1) \quad m_t - p_t = -\lambda \pi_t^e$$

$$\Leftrightarrow \pi_t^e = -\frac{1}{\lambda} (m_t - p_t) \quad \forall t$$

$$\text{in } t-1 \quad \pi_{t-1}^e = -\frac{1}{\lambda} (m_{t-1} - p_{t-1})$$

plug into (2)

$$\begin{aligned} \Leftrightarrow \pi_t^e &= \lambda \pi_{t-1}^e + (1-\lambda)(p_t - p_{t-1}) \\ -\frac{1}{\lambda} (m_t - p_t) &= -\frac{\lambda}{\lambda} (m_{t-1} - p_{t-1}) + (1-\lambda)(p_t - p_{t-1}) \end{aligned}$$

$\times f(x):$

$$m_t - p_t = \lambda(m_{t-1} - p_{t-1}) - \alpha(1-\lambda)(p_t - p_{t-1})$$

$$p_t(-1 + \alpha(1-\lambda)) = -m_t + \lambda m_{t-1} + (-\lambda + \alpha(1-\lambda))p_{t-1}$$

$$p_t = \underbrace{\frac{\lambda - \alpha(1-\lambda)}{1 - \alpha(1-\lambda)}}_{\beta_1} p_{t-1} + \underbrace{\frac{m_t}{1 - \alpha(1-\lambda)}}_{\beta_2} + \underbrace{\frac{-\lambda}{1 - \alpha(1-\lambda)}}_{\beta_3} m_{t-1}$$

Solution

$$p_t = \frac{1 - \alpha(1-\lambda)}{1 - \alpha(1-\lambda)} p_{t-1}$$

$$+ \frac{1}{1 - \alpha(1-\lambda)} m_t + \frac{-\lambda}{1 - \alpha(1-\lambda)} m_{t-1}$$

Big question

Can we have hyperinflation when money supply m_t does not explode?

Math

$$x_t = p x_{t-1} + \underline{b}$$

• if $p > 1$

• if $0 < p < 1$

